

POLICY AND GUIDELINES FOR CAPSTONE PROJECT AND RESEARCH 1

I. Policy Statement

The Capstone Project and Research 1 shall serve as a culminating academic requirement designed to develop, assess, and demonstrate the students' ability to apply knowledge, skills, and competencies acquired in their program of study. It shall provide students with an opportunity to identify real-world problems, propose technology-based or research-based solutions, conduct systematic inquiry, develop appropriate systems or outputs, and present their work through formal documentation and oral defense.

This policy establishes the official guidelines, schedule, group requirements, documentation format, defense procedures, expected outputs, and academic standards for the conduct of Capstone Project and Research 1. All students enrolled in the course shall comply with these guidelines to ensure uniformity, fairness, academic integrity, and quality in the preparation, development, presentation, and evaluation of their capstone or research outputs.

The Capstone Project and Research 1 shall emphasize problem identification, conceptualization, research writing, technical planning, system design, feasibility assessment, and progressive development. Students are expected to produce a capstone project or research output that is relevant, feasible, ethical, technically sound, and beneficial to the target users, institution, community, or industry.

II. Objectives of the Policy

This policy aims to:

1. Provide a clear and organized framework for the conduct of Capstone Project and Research 1.
2. Guide students in the preparation, documentation, development, and defense of their capstone or research project.
3. Establish uniform requirements for grouping, scheduling, documentation, and presentation.
4. Ensure that proposed projects are feasible, relevant, innovative, and aligned with program outcomes.
5. Promote academic honesty, originality, collaboration, professionalism, and accountability.
6. Define the expected outputs and completion requirements for each stage of the capstone and research process.

7. Set standards for evaluating students' progress from concept development to final project completion.

III. Scope and Coverage

This policy shall apply to all students enrolled in Capstone Project and Research I. It shall cover the following activities:

1. Orientation on capstone and research requirements;
2. Formation of project groups;
3. Conceptualization and design thinking process;
4. Preparation and submission of documentation requirements;
5. Concept defense;
6. Title defense;
7. Progress reports;
8. Final defense;
9. System development or project implementation;
10. Submission of required documents and compliance reports.

All students, group members, course teachers, advisers, and panel members involved in the conduct of Capstone Project and Research I shall be guided by this policy.

IV. General Guidelines

A. Student Responsibility

Students shall be responsible for the planning, preparation, development, documentation, and presentation of their capstone or research project. They shall ensure that their work is original, properly documented, ethically conducted, and submitted within the prescribed deadlines.

Each student shall actively participate in all stages of the project. Non-participation, repeated absences, failure to contribute, or negligence in performing assigned tasks may be subject to appropriate academic action as determined by the course teacher.

B. Role of the Course Teacher

The course teacher shall supervise the overall conduct of Capstone Project and Research I. The course teacher shall provide orientation, monitor group formation, facilitate discussions on documentation requirements, guide students in conceptualization, determine the feasibility of proposed projects during concept defense, and coordinate the conduct of succeeding defense activities.

The course teacher shall also have the authority to require revisions, reject infeasible proposals, recommend regrouping when necessary, and enforce compliance with academic standards and submission deadlines.

C. Academic Integrity

All capstone and research outputs must be original. Plagiarism, fabrication of data, misrepresentation of results, unauthorized use of existing systems, and other forms of academic dishonesty are strictly prohibited.

Students must properly cite all sources used in their documentation, including books, journal articles, websites, existing systems, software tools, frameworks, datasets, and other references. Any violation of academic integrity shall be subject to institutional policies and appropriate academic sanctions.

V. Grouping Guidelines

Students shall form groups composed of a minimum of three members and a maximum of four members.

Each group shall be responsible for assigning internal roles and responsibilities to ensure efficient completion of tasks. Recommended roles may include, but are not limited to:

1. Project Leader;
2. Documentation Lead;
3. System Developer or Technical Lead;
4. Research and Data Lead;
5. Quality Assurance and Compliance Lead.

Although roles may be assigned, all members shall remain collectively responsible for the quality and completion of the project. The failure of one member to perform assigned tasks shall not exempt the group from meeting course requirements and deadlines.

Changes in group membership shall only be allowed upon approval of the course teacher and must be supported by valid reasons.

VI. Schedule of Activities

The following schedule shall guide the conduct of Capstone Project and Research 1:

A. June 3 – Orientation of the Capstone Project

An orientation shall be conducted to discuss the nature, objectives, requirements, expected outputs, grading considerations, documentation format, defense procedures, and timeline of the Capstone Project and Research 1.

Students shall be informed of their responsibilities, grouping requirements, and the standards expected throughout the course.

B. June 3–5 – Determination of Group Members

Students shall form their respective groups within the prescribed period. Each group must comply with the required number of members, which is a minimum of three and a maximum of four.

The final list of group members shall be submitted to the course teacher for approval.

C. June 8–12 – Conceptualization, Design Thinking Process, and Documentation Discussion

During this period, students shall engage in conceptualization activities and apply the design thinking process to identify possible problems, users, needs, and solutions.

The course teacher shall also present and discuss the required documentation format and content. Students shall be guided on how to prepare the necessary sections of the capstone or research manuscript.

D. June 15–19 – Concept Defense

The Concept Defense shall be conducted in the classroom. During this stage, each group shall present its proposed concept to the course teacher.

The course teacher shall determine the feasibility, relevance, originality, and technical soundness of the proposed capstone project. The proposed concept may be approved, approved with revisions, deferred for improvement, or rejected if found infeasible or inappropriate.

E. June 22–26 – Title Defense

The Title Defense shall be conducted to evaluate the proposed project title and the initial manuscript covering Chapters 1 to 3.

During this stage, the group shall demonstrate that the proposed project has a clear problem, defined objectives, established scope, adequate related literature and systems, and appropriate technical background.

F. June 29–July 3 – Submission of Required Documents

Groups shall submit the required documents based on the prescribed format and content. Late, incomplete, or non-compliant submissions may be subject to penalties as determined by the course teacher.

VII. Manner and Stages of Defense

The Capstone Project and Research 1 shall follow a progressive defense structure. Each defense stage shall evaluate the development, documentation, feasibility, and quality of the project.

The required stages are as follows:

1. Concept Defense;
2. Title Defense;
3. Progress Report 1;
4. Progress Report 2;
5. Final Defense.

Each stage must be completed before the group may proceed to the next major requirement, unless otherwise permitted by the course teacher.

VIII. Requirements per Defense Stage

A. Concept Defense

For the Concept Defense, each group shall present the following:

1. Project concept;
2. Problem setting;
3. Proposed solution;
4. Technical background.

The purpose of the Concept Defense is to determine whether the proposed project is relevant, feasible, technically appropriate, and aligned with the course requirements.

The presentation must clearly answer the following questions:

1. What problem does the project intend to address?
2. Who are the target users or beneficiaries?
3. Why is the problem important?
4. What solution is being proposed?
5. What technologies, tools, platforms, or methods may be used?
6. Is the project feasible within the given time, resources, and student capability?

The course teacher shall provide feedback and may require revisions before allowing the group to proceed to the Title Defense.

B. Title Defense

For the Title Defense, each group shall submit and present Chapters 1 to 3 of the manuscript.

The required coverage shall include:

1. Chapter 1 – Introduction;
2. Chapter 2 – Review of Related Literature and System;
3. Chapter 3 – Technical Background.

The Title Defense shall determine whether the proposed title, problem, objectives, scope, literature, related systems, and technical requirements are properly established.

The panel or course teacher may recommend modification of the title, refinement of objectives, improvement of the scope and limitations, addition of related studies, revision of technical requirements, or restructuring of the documentation.

C. Progress Report 1

For Progress Report 1, each group shall submit and present Chapters 1 to 4 of the manuscript and demonstrate a 60% working system.

The required output shall include:

1. Updated Chapter 1 – Introduction;
2. Updated Chapter 2 – Review of Related Literature and System;
3. Updated Chapter 3 – Technical Background;
4. Chapter 4 – Methodology;
5. Demonstration of at least 60% working system functionality.

The group must show substantial progress in both documentation and system development. The 60% working system should already reflect the core features, initial interface, database structure if applicable, and basic operational flow of the proposed solution.

D. Progress Report 2

For Progress Report 2, each group shall submit and present Chapters 1 to 5 of the manuscript and demonstrate a 90% working system.

The required output shall include:

1. Updated Chapters 1 to 4;
2. Chapter 5 – Results and Discussions;
3. Demonstration of at least 90% working system functionality.

At this stage, the system should be nearly complete. Most major features must already be functional, tested, and aligned with the objectives of the project. The group shall also present initial results, findings, testing outcomes, user feedback if applicable, and discussion of system performance.

E. Final Defense

For the Final Defense, each group shall submit a softbound copy of the complete manuscript and present a 100% working system with a compliance report.

The required output shall include:

1. Complete softbound manuscript;
2. Complete Chapters and required sections;
3. Fully functional 100% working system;
4. Compliance report showing how previous comments, recommendations, and required revisions were addressed;
5. Final project presentation;
6. System demonstration.

The Final Defense shall determine whether the project meets the required academic, technical, documentation, and presentation standards. The project may be accepted, accepted with minor revisions, accepted with major revisions, or required for re-defense depending on the evaluation results.

IX. Required Manuscript Sections

The capstone or research manuscript shall follow the prescribed sections below.

Chapter 1 – Introduction

A. Project Context

This section shall present the background of the study or project. It shall describe the situation, issue, problem, or opportunity that led to the development of the proposed capstone or research project.

The Project Context must establish the relevance and importance of the study.

B. Purpose and Description

This section shall explain the purpose of the project and provide a clear description of the proposed system, study, or output.

It shall answer what the project intends to do and how it will address the identified problem.

C. Objectives

This section shall present the general and specific objectives of the project.

The objectives must be clear, measurable, attainable, relevant, and aligned with the proposed solution.

D. Scope and Limitation

This section shall define the coverage and boundaries of the project. It shall specify what the project will include and what it will not include.

Limitations may include technical constraints, time constraints, user coverage, data limitations, platform limitations, or resource limitations.

E. Definition of Terms

This section shall define important terms, technical words, concepts, and variables used in the study.

Terms must be defined operationally, especially when they have specific meaning within the project.

Chapter 2 – Review of Related Literature and System

A. Related Literature

This section shall present published sources, studies, articles, books, journals, and other references that support the project.

The literature must help establish the background, relevance, and theoretical or conceptual basis of the study.

B. Related System

This section shall discuss existing systems, applications, platforms, or solutions related to the proposed project.

The discussion must include the features, strengths, weaknesses, and relevance of each related system.

C. Synthesis of the State-of-the-Art

This section shall summarize and analyze the reviewed literature and systems. It shall identify current trends, common features, limitations, and best practices.

The synthesis must show how the reviewed works contributed to the development of the proposed project.

D. Gap Bridged by the Study

This section shall identify the specific gap, limitation, or unresolved issue found in existing literature or systems that the proposed project intends to address.

The group must clearly explain how their project offers an improvement, solution, or contribution.

Chapter 3 – Technical Background

A. Software Requirements

This section shall identify and describe the software tools, platforms, programming languages, frameworks, database systems, libraries, and applications needed to develop and run the project.

The group must explain why the selected software tools are appropriate.

B. Hardware Requirements

This section shall identify the hardware specifications, devices, equipment, or technical infrastructure required for the development, testing, deployment, or use of the project.

The hardware requirements must be realistic and available within the project's resources.

Chapter 4 – Methodology

A. Sources of Data

This section shall identify where the data or information needed for the project will come from.

Sources may include respondents, users, documents, institutional records, interviews, surveys, observations, system logs, datasets, or other relevant sources.

B. Data Instrumentation

This section shall describe the tools, instruments, or methods used to gather data.

Examples may include questionnaires, interview guides, observation checklists, usability testing forms, evaluation rubrics, and other research instruments.

The methodology must also describe the development approach, process model, or design method used in the project when applicable.

Chapter 5 – Results and Discussions

This section shall present the results of the project development, testing, evaluation, and implementation.

It shall include a discussion of the system features, testing results, user evaluation, performance, effectiveness, and alignment with project objectives.

Tables, figures, screenshots, charts, and other supporting materials may be included when appropriate.

Chapter 6 – Conclusions and Recommendations

A. Conclusions

This section shall present the conclusions based on the results of the project.

The conclusions must directly answer the objectives of the study and must be supported by findings.

B. Recommendations

This section shall present recommendations for improvement, future development, implementation, maintenance, or further research.

Recommendations must be realistic and based on the limitations and findings of the project.

X. Documentation Guidelines

All manuscripts must follow the format prescribed by the course teacher. Students shall ensure consistency in formatting, grammar, citation style, pagination, heading structure, tables, figures, and references.

The manuscript must be written in formal academic language. Informal expressions, unsupported claims, vague statements, and poorly cited information shall be avoided.

All figures, diagrams, tables, screenshots, and appendices must be properly labeled and discussed in the manuscript.

Revisions required by the course teacher or panel must be incorporated before the next defense or submission deadline.

XI. System Development Guidelines

For capstone projects involving system development, the system must be aligned with the approved title, objectives, scope, and methodology.

The system must be developed progressively according to the required completion percentage per stage:

1. Progress Report 1 – 60% working system;
2. Progress Report 2 – 90% working system;
3. Final Defense – 100% working system.

The system shall be evaluated based on functionality, usability, reliability, relevance, completeness, technical quality, and compliance with approved requirements.

A system shall not be considered complete if major features are missing, core functions are not working, or the output does not address the approved objectives.

XII. Compliance Report

A compliance report shall be required during the Final Defense. The compliance report must document the comments, suggestions, corrections, and recommendations given during previous defense stages and explain how the group addressed each item.

The compliance report shall serve as proof that the group has considered and implemented the required revisions.

The report may include the following columns:

1. Comment or recommendation;
2. Source of comment;
3. Action taken;
4. Page number or system module affected;
5. Status of compliance.

XIII. Attendance and Participation

All group members are required to attend scheduled orientations, consultations, presentations, and defenses.

Absence during defense or major presentation shall only be excused for valid reasons, subject to the approval of the course teacher. A student who fails to attend a required defense without valid reason may receive an individual penalty or may be required to complete additional requirements.

Each member must be able to explain the project, documentation, and assigned tasks. During defense, panel members or the course teacher may ask questions to any member of the group.

XIV. Submission Policy

All required documents must be submitted on or before the prescribed deadline. Submissions must follow the format, file type, naming convention, and method of submission required by the course teacher.

Late submissions may be subject to deductions or may not be accepted unless a valid reason is presented and approved.

Incomplete submissions shall be considered non-compliant and may affect the group's standing in the course.

XV. Consultation and Monitoring

Groups are encouraged to consult regularly with the course teacher regarding their project concept, documentation, methodology, technical design, and system development.

The course teacher may require progress checks, draft submissions, system demonstrations, or consultation logs to monitor the group's development.

Failure to consult or show progress may affect the group's eligibility to proceed to the next defense stage.

XVI. Evaluation Guidelines

The evaluation of the Capstone Project and Research I may include the following criteria:

1. Relevance and significance of the problem;
2. Clarity of objectives;
3. Feasibility of the proposed solution;
4. Quality of documentation;
5. Adequacy of literature and related system review;
6. Soundness of technical background;
7. Appropriateness of methodology;
8. Progress and completeness of system development;
9. Quality of presentation and defense;
10. Ability of members to answer questions;
11. Originality and academic integrity;
12. Compliance with comments and recommendations.

The course teacher may provide a detailed rubric for each defense stage.

XVII. Grounds for Revision, Rejection, or Re-Defense

A group may be required to revise, change, or defend again under any of the following circumstances:

1. The proposed project is not feasible within the given time or resources;
2. The project does not address a clear problem;
3. The title and objectives are not aligned;
4. The manuscript is incomplete or poorly prepared;
5. The system output does not match the approved proposal;
6. The system completion percentage does not meet the required level;
7. The group fails to comply with previous comments and recommendations;
8. The project contains plagiarized, fabricated, or misrepresented content;
9. The group fails to demonstrate sufficient understanding of the project;
10. Major technical or documentation issues remain unresolved.

XVIII. Ethical Considerations

Projects involving users, respondents, personal information, institutional records, or sensitive data must observe ethical standards.

Students shall ensure confidentiality, informed consent, proper data handling, and responsible use of information. Personal data must not be collected, stored, or disclosed without proper permission.

Systems that involve user accounts, records, transactions, or personal information must include appropriate security and privacy considerations.

XIX. Final Output Requirements

For successful completion of Capstone Project and Research 1, each group must submit and present the required outputs based on the defense stage.

The final expected outputs shall include:

1. Approved project title;
2. Complete manuscript;
3. Complete and functional system or project output;
4. Required documentation;
5. Compliance report;
6. Presentation materials;
7. Other supporting documents required by the course teacher.

XX. Effectivity

This policy shall take effect upon approval and implementation by the course teacher, department, or academic unit concerned. It shall remain in force for the duration of Capstone Project and Research 1 unless revised, amended, or superseded by subsequent official guidelines.

All students enrolled in the course are expected to read, understand, and comply with this policy. Failure to comply with the provisions stated herein may result in academic consequences based on the rules and regulations of the course and the institution.

XXI. Concluding Provision

The Capstone Project and Research I is a major academic undertaking that requires discipline, teamwork, creativity, technical competence, research skills, and professionalism. Through this policy, students are expected to produce meaningful, ethical, feasible, and high-quality outputs that demonstrate their readiness to apply their knowledge and skills in solving real-world problems.

Strict compliance with these guidelines shall ensure a systematic, fair, and productive conduct of the course and shall contribute to the successful completion of each group's capstone or research project.